

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES				
10							***** MODULAR VALVES *****			812	NIPPLE TO BE CUT FROM SS SCH XXS PIPE.			
20										815	TO BE USED WITH COMPACT FLANGES ONLY.			
20	U04		MODULAR DBB	K15AC02	0.5	1.5	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	816	COMPACT FLANGES CAN BE USED FOR TIE-IN PURPOSE ONLY TO EXISTING PIPING AND EQUIPMENT.			
30	U04		MODULAR DBB	K15AC06	2	2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	817	USE ONLY FOR TIE IN			
40	U04	VAV	MODULAR DBB	K09AC06	3	3	MODULAR VALVE DB&B BALL TYPE 900# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO FULL SIZE TYPE 2BV	0	0	B1	ALL BOLTS ARE METRIC DIA.			
50	U04		MODULAR DBB	M15AC02	0.75	1.5	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	B8	3/4" X 130MM BOLTS ONLY TO BE USED FOR 3/4 INCH 1500# PERAR BALL VALVES ITEM CODE B15AC02			
60	U04		MODULAR DBB	M15AC06	2	2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	BA1	TO BE USED WITH RAISED FACE FLANGES ONLY.			
70	U04	VAW	MODULAR DBB	M15EC02	0.75	1	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO REDUCING - CONTROLLED BORE TYPE R2BVR	2	2	FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.			
80	U04	VAW	MODULAR DBB	M09EC02	1	1	MODULAR VALVE DB&B BALL TYPE 900# X 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3	3	GA6	SPECIAL GASKET ONLY FOR USE AS SPECIFIED IN ENG-200-117			
90							***** BALL VALVES *****			UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.			
90										VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.			
100	U04	B8	BALL	B15AC02	0.75	0.75	BALL 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM FLOATING METAL FB LO	0	0	VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.			
110							***** GATE VALVES *****			VC1	SPRING PISTON CHECK VALVES ARE PREFERRED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.			
120	U04		GATE	G15AC48	0.75	1.5	GATE 1500# RF LTCS A350 LF2 CL.1 TRIM 16 WEDGE HW	0	0	ZA8	*****TO BE USED IN STEAM AND CONDENSATE SERVICE ONLY*****			
130			GATE	V15AC41	2	2	GATE PARALLEL SLIDE 1500# RF LTCS A352 LCC TRIM 16 HW	0	0					
140			GATE	V09AC41	3	4	GATE PARALLEL SLIDE 900# RF LTCS A352 LCC TRIM 16 HW	0	0					
150			GATE	V09AC42	6	8	GATE PARALLEL SLIDE 900# RF LTCS A352 LCC TRIM 16 GO	0	0					
160							***** BUTTERFLY VALVES *****							
160														
170		ZA8	BUTTERFLY	T09AC61	10	10	BUTTERFLY 900# RF LTCS A352 LCC INC TRIM TOV METAL DBL FLG SHORT PAT GO	0	0					
180		ZA8	BUTTERFLY	T09CC61	12	24	BUTTERFLY 900# COMPACT FLANGE LTCS A352 LCC INC TRIM TOV METAL DBL FLG SHORT PAT GO	0	0					
190							***** GLOBE VALVES *****							
190														
200														
200	U04		GLOBE	L15AC48	0.75	1.5	GLOBE 1500# RF LTCS A350 LF2 CL.1 INCONEL OS&Y HW	0	0					
210	U04		GLOBE	L15AC41	2	2	GLOBE 1500# RF LTCS A352 LCC INCONEL OS&Y HW	0	0					
220			GLOBE	L09AC41	3	3	GLOBE 900# RF LTCS A352 LCC TRIM 16 OS&Y HW	0	0					
230			GLOBE	L09AC42	4	8	GLOBE 900# RF LTCS A352 LCC TRIM 16 OS&Y GO	0	0					
240							***** CHECK VALVES *****							
240														
250														
250	U04	VC1	CHECK	C15AC50	0.75	1.5	CHECK PISTON 1500# RF LTCS A350 LF2 CL.1 TRIM 16	0	0					
260	U04		CHECK	C15AC45	2	2	CHECK DUAL WAFER LUG 1500# RF LTCS A352 LCC INCONEL TRIM THROU BOLTED	0	0					
270			CHECK	C09AC45	3	8	CHECK DUAL WAFER LUG 900# RF LTCS A352 LCC TRIM 16 THROU BOLTED	0	0					
280			CHECK	C09AC52	10	24	CHECK DUAL DBL FLG 900# RF LTCS A352 LCC TRIM 16	0	0					
290							***** PIPE *****							
290														
300														
300			PIPE	5358512	0.75	1.5	PIPE SCH 160 SMLS LTCS A333 GR.6 PE	0	0					
310			PIPE	5358515	2	3	PIPE SCH 160 SMLS LTCS A333 GR.6 BE	0	0					
320			PIPE	5358510	4	16	PIPE SCH 120 SMLS LTCS A333 GR.6 BE	0	0					
330			PIPE	5358600	18	24	PIPE SCH 120 EFW LTCS A671 C060 CL.22 BE	0	0					
340		812	PIPE	5358511	0.75	0.75	PIPE SCH XXS SMLS LTCS A333 GR.6 PE	0	0					
350							***** FITTINGS *****							
350														
360														
360			ELBOW	5392816	0.75	3	ELL BW 90 DEG LR SCH 160 LTCS A420 WPL6	0	0					
370			ELBOW	5392700	4	16	ELL 90 DEG LR SCH 120 LTCS A420 WPL6	0	0					
380			ELBOW	5392388	18	24	ELL 90 DEG LR SCH 120 LTCS A420 WPL6-W	0	0					
390			ELBOW	5392822	0.75	3	ELL BW 45 DEG LR SCH 160 LTCS A420 WPL6	0	0					
400			ELBOW	5392701	4	16	ELL BW 45 DEG LR SCH 120 LTCS A420 WPL6	0	0					
410			ELBOW	5392392	18	24	ELL 45 DEG LR SCH 120 LTCS A420 WPL6-W	0	0					
420														
420			TEE	5392825	0.75	3	TEE BW SCH 160 LTCS A420 WPL6	0	0					
430			TEE	5392702	4	16	TEE BW SCH 120 LTCS A420 WPL6	0	0					
440			TEE	5392404	18	24	TEE BW SCH 120 LTCS A420 WPL6-W	0	0					
450			TEE RED	5392821	0.75	2	TEE RED BW SCH 160 LTCS A420 WPL6	1	3					
460			TEE RED	5392470	1.5	3	TEE RED SCH 120/160 LTCS A420 WPL6	4	6					
470			TEE RED	5392603	4	14	TEE RED SCH 120 LTCS A420 WPL6	6	16					
480			TEE RED	5392878	8	20	TEE RED SCH 120 LTCS A420 WPL6-W	18	24					
490														
490			CAP	5392826	0.75	3	CAP BW SCH 160 LTCS A420 WPL6	0	0					
			TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN		CONSTRUCTION		CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
					ALL SIZES - BUTT WELD PWHT FOR WT >19 MM SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE		CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 900 RF/CF TEMPERATURE RANGE: -40 / 410 °C	12/16/2024	BY OZAC	U04	Page 1 of 2	900H25

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500			CAP	5392703	4	24	CAP SCH 120 LTCS A420 WPL6	0	0	
510										
510			RED CONC	5392827	0.75	2	REDUCER CONC SCH 160 LTCS A420 WPL6	1	3	
520			RED CONC	5392467	1.5	3	REDUCER CONC SCH 120/160 LTCS A420 WPL6	4	6	
530			RED CONC	5392704	4	14	REDUCER CONC SCH 120 LTCS A420 WPL6	6	16	
540			RED CONC	5392707	10	20	REDUCER CONC SCH 120 LTCS A420 WPL6-W	18	24	
550			RED ECC	5392828	0.75	2	REDUCER ECC SCH 160 LTCS A420 WPL6	1	3	
560			RED ECC	5392468	1.5	3	REDUCER ECC SCH 120/160 LTCS A420 WPL6	4	6	
570			RED ECC	5392705	4	14	REDUCER ECC SCH 120 LTCS A420 WPL6	6	16	
580			RED ECC	5392709	10	20	REDUCER ECC SCH 120 LTCS A420 WPL6-W	18	24	
590							***** OLETS *****			
600										
600	U04		OLETS	5504170	0.75	2	NIPOFLANGE 1500# RF SCH 160 LTCS A350 LF2 CL.1	2	3	
610	U04		OLETS	5504171	0.75	2	NIPOFLANGE 1500# RF SCH 120/160 LTCS A350 LF2 CL.1	4	24	
620	U04	UJ2	OLETS	5504163	1.5	1.5	NIPOFLANGE 1500# RF SCH 120/160 LTCS A350 LF2 CL.1	4	24	
630										
630	U04		OLETS	5505392	0.75	1	WELDOLET SCH 160 LTCS A350 LF2 CL.1	2	3	
640	U04		OLETS	5505394	0.75	3	WELDOLET SCH 120/160 LTCS A350 LF2 CL.1	4	24	
650	U04		OLETS	5505195	4	8	WELDOLET SCH 120 LTCS A350 LF2 CL.1	12	24	
660	U04		OLETS	5504379	1	1.5	WELDOLET FLAT BOTTOM SCH 160 LTCS A350 LF2 CL.1	0	0	
670										
670	U04	UJ2	ELBOFLG	5516082	1.5	1.5	ELBOFLANGE 1500# RF SCH 120/160 LTCS A350 LF2 CL.1	4	24	
680										
680		816					***** FLANGES *****			
690										
690	U04	FA8	TEE RED	5599704	0.5	0.5	FLG SCR D (FNPT) REDUCING 1500# RF LTCS A350 LF2 CL.1	0.75	1.5	
700	U04		FLANGE	5557397	0.75	2	FLG WN 1500# RF LTCS A350 LF2 CL.1 SCH 160 BORE	0	0	
710	U04		FLANGE	5557145	3	3	FLG WN 900# RF LTCS A350 LF2 CL.1 SCH 160	0	0	
720	U04	FA8	FLANGE	5557143	4	24	FLG WN 900# RF LTCS A350 LF2 CL.1 SCH 120 BORE	0	0	
730	U04	817	FLANGE	5619821	12	24	FLG COMPACT 900# RJ LTCS A350 LF2 CL.1 SCH 120 BORE	0	0	
740	U04		BLIND	5615616	0.5	2	FLG BLIND 1500# RF LTCS A350 LF2 CL.1	0	0	
750	U04	FA8	BLIND	5615605	3	24	FLG BLIND 900# RF LTCS A350 LF2 CL.1	0	0	
760	U04	817	BLIND	5619771	12	24	FLG COMPACT BLIND 900# RJ LTCS A350 LF2 CL.1 PIPE CLASS 900H25	0	0	
770										
770							***** ORIFICE FLANGES *****			
780										
780	U04		ORIFICE FLG	5624598	2	2	WN ORIF SET 1500# RF LTCS A350 LF2 CL.1 SCH 160 BORE	0	0	
790			ORIFICE FLG	5624537	3	3	WN ORIF SET 900# RF LTCS A350 SCH 160 BORE	0	0	
800			ORIFICE FLG	5624532	4	10	WN ORIF SET 900# RF LTCS A350 SCH 120 BORE	0	0	
810										
810							***** PIPE NIPPLES *****			
820										
820		812		0			NIPPLE BBE - CUT FROM PIPE			
830										
830							***** GASKETS *****			
840										
840			GASKET	5672251	0.5	2	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
850			GASKET	5673117	3	24	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
860	U04	817	GASKET	5673441	12	24	GASKET 900# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 900H25	0	0	
870										
870		B1					***** BOLTING *****			
880										
880	U04	815	BOLTING	5675770	0.5	30	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
890		BA1	BOLTING	5675776	24	99	STUD-BOLT WITH 2 NUTS API-6A718	0	0	
900										
900	U04	GA6	GASKET	5673021	2	2	GASKET 1500# 316SS KAMMPROFILE GRAPHITE ASME B16.5	0	0	
910	U04	GA6	GASKET	5673020	3	3	GASKET 900# 316SS KAMMPROFILE GRAPHITE ASME B16.5	0	0	
920										
920							***** BLINDS AND SPACERS *****			
930										
930			SPEC BLIND	5749595	0.75	2	SPECTACLE BLIND 1500# RF LTCS A516 GR.70	0	0	
940			SPEC BLIND	5749596	3	8	SPECTACLE BLIND 900# RF LTCS A516 GR.70	0	0	
950										
950			PADDLE	5750102	10	24	PADDLE BLANK 900# RF LTCS A516 GR.70	0	0	
960			PADDLE	5750202	10	24	PADDLE SPACER 900# RF LTCS A516 GR.70	0	0	
970	U04	817	PADDLE	5750215	12	24	PADDLE BLANK COMPACT 900# RJ LTCS A516 GR.70	0	0	
980	U04	817	PADDLE	5750223	12	24	PADDLE SPACER COMPACT 900# RJ LTCS A516 GR.70	0	0	
990										

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
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